

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: Engineering

Branch: Common To All Branches

Semester : 1st Sem

Subject Code & Name: 24AF2CMEES108 (Basic Civil and Mechanical Engineering)

Max Marks: 60

Date: 22/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)	Remember	12
1	A..... is a horizontal member which is placed across an opening to support the position of the structure above it.		1
	a. Doors b. Windows c. Sill d. Lintel		
2	what is the primary function of a shallow foundation		1
	a. To transfer loads to a deeper b. To distribute loads over a larger area to reduce soil pressure c. To provide Lateral Support to the superstructure d. To resist uplift forces		
3	What is the primary function of cement in concrete?		1
	a. To provide Strength b. To improve workability c. To reduce shrinkage d. To increase durability		
4	The entire assembly of styles, Pannels & rails is known as the.....		1
	a Putty b. Horn c. Sash d. Shutter		
5	What is surveying?		1
	a. To find elevations w.r to datum b. Show's the relative positions of the object on the c. To find elevations of points having d. All of above		

		surface of the earth	same contour intervals			
6	What is the term used for an imaginary line on the ground Joining of equal elevations					1
	a. Level Line	b. Line of sight	c. Datum	d. Contour		
7	Which of the following kind of energy output is obtained from a 'Steam Power Plant'					1
	a. Electricity	b. Thermal energy	c. Sound energy	d. Heat energy		
8	Which of the following is a classification of automobiles based on Load?					1
	a. Heavy transport vehicle (HTV)	b. Sedan Hatchback car	c. Four wheeler vehicle	d. Front-wheel drive		
9	Petrol engines are..... than diesel engines.					1
	a. Lighter	b. Unpredictable	c. Heavier	d. None of the mentioned		
10	Which of the following is a type of thermodynamic system?					1
	a. Open system	b. Closed system	c. Thermally isolated system	d. All of the mentioned		
11	Sand mold and permanent mold are the parts of..... manufacturing process.					1
	a. Machining	b. Casting	c. Welding	d. None of above		
12	Which of the following operation is not carried out on Lathe Machine.					1
	a. Welding	b. Facing	c. Turning	d. Parting		
Q. 2	Solve the following.				CO2	12
A)	Describe in detail concrete with its types & properties?					6
B)	Explain the role of Civil engineer in the field of Construction engineering?					6

Q.3	Solve the following.	CO3	12
A)	Explain the foundation with its types & function of foundation?		6
B)	Describe in detail 'lean to roof' with sketch?		6
Q. 4	Solve Any Two of the following.	CO4	12
A)	Describe in detail "Metric chain" with sketch?		6
B)	Describe Advantages & Disadvantages of "Plane Table surveying"?		6
C)	Define the terms? i) Reduced level ii) Height of instrument iii) Contour line iv) Bench marks v) Change Point vi) Contour interval		6
Q.5	Solve Any Two of the following.	CO5	12
A)	Describe first law of thermodynamics. Enlist the limitations of the same?		6
B)	Classify Internal combustion engines based on (i) Cycle of operation, ii) Fuel used, iii) Cylinder Arrangement, (iv) No. of strokes, (v) Application, (vi) Ignition method.		6
C)	What is the function of Power plant. Explain in brief working of thermal power plant with suitable sketch?		6
Q. 6	Solve Any Two of the following.	CO5	12
A)	What is the difference between machine and mechanism? Explain any two types of mechanism with suitable diagrams.		6
B)	Classify engineering materials. Describe in detail properties and applications of any two non-ferrous metals?		6
C)	With suitable diagrams describe any six operations performed on Lathe machine?		6
	*** End ***		